

CHAPPIDI RAMA PRAKASH

Full Stack Developer | Backend Engineer | Cloud & Real-Time Systems

+91 9390833979 | chappidiramaprakash167@gmail.com | github.com/ram294167 | Amalapuram, Andhra Pradesh

PROFESSIONAL SUMMARY

Results-driven Full Stack / Backend Developer with hands-on experience designing and deploying scalable, production-grade systems. Proficient in building RESTful APIs, real-time communication platforms, and cloud-hosted services using Node.js, Express.js, PostgreSQL, Redis, Socket.IO, and AWS. Demonstrated ability to architect end-to-end solutions — from database schema design to EC2 deployment — for service booking, live tracking, and subscription-based products. Passionate about clean architecture, performance optimization, and delivering reliable systems at scale.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python
Backend	Node.js, Express.js, REST APIs, MVC Architecture, Role-Based Access Control (RBAC), JWT Authentication, Middleware Design
Real-Time	Socket.IO, Live GPS Tracking, Bidirectional Communication, Event-Driven Architecture
Databases	PostgreSQL, Redis — Schema Design, Query Optimization, Indexing, Caching, Session Management, Stored Procedures, Triggers
Cloud & DevOps	AWS EC2 (instance setup, SSH, security groups), AWS S3 (media storage, presigned URLs), Nginx (reverse proxy, SSL), PM2 (process management)
Tools	Git, GitHub, Postman, Linux CLI, VS Code
Integrations	Payment Gateway APIs, Third-Party Service APIs, Push Notifications, Webhook Handlers

PROFESSIONAL EXPERIENCE

Database Intern • [Algonox Pvt. Ltd.](#)

Oct 2024 – Jan 2025

Database Management, Migration & Backend Integration

- ▶ Executed a full-scale data migration from Microsoft SQL Server (MSSQL) to PostgreSQL, covering schema translation, data type mapping, and post-migration validation — ensuring zero data loss and consistent query performance across both systems.
- ▶ Designed and authored SQL scripts, stored procedures, and triggers to automate repetitive data operations, reducing manual overhead for backend teams by standardizing data workflows.
- ▶ Participated in schema normalization and relational database design to support scalable, high-performance backend services — applying principles of 3NF to minimize redundancy.
- ▶ Collaborated closely with backend developers to integrate optimized database layers with RESTful API endpoints, improving response times and system maintainability.
- ▶ Conducted performance profiling of slow-running queries using EXPLAIN ANALYZE and applied indexing strategies to improve query execution plans.

PROJECTS

CoreFix MEP Services – Service Management Platform

Jan 2026 – Present

Tech Stack: Node.js, Express.js, PostgreSQL, AWS EC2, AWS S3, JWT, REST APIs

- ▶ Architected and developed a multi-module backend system for a Mechanical, Electrical & Plumbing (MEP) service company — handling service booking, AMC (Annual Maintenance Contract) subscription management, payment workflows, user roles, and technician job assignment.
- ▶ Designed normalized PostgreSQL schemas supporting complex relationships between customers, service requests, technician profiles, job statuses, and billing records, enabling efficient queries across all modules.
- ▶ Built secure REST API endpoints with JWT-based authentication and role-based access control (RBAC) separating Admin, Technician, and Customer permissions — preventing unauthorized data access across all routes.
- ▶ Deployed and configured production infrastructure on AWS EC2: set up Ubuntu server, configured Nginx as a reverse proxy, secured with SSL, and managed Node.js processes using PM2 for zero-downtime restarts.
- ▶ Integrated AWS S3 for scalable media storage — enabling secure upload and retrieval of service reports, job photos, and user documents using pre-signed URLs with time-limited access control.
- ▶ Implemented structured payment workflow APIs to handle service billing, subscription renewal reminders, and payment status tracking, integrating with third-party payment providers.

Vegaa – Real-Time Service Booking Platform

Mar 2026 – Present

Tech Stack: Node.js, Express.js, Socket.IO, Redis, PostgreSQL, AWS EC2, Nginx

- ▶ Designed and built a real-time on-demand service booking platform inspired by Uber/Rapido — connecting users with nearby service providers with live status updates and location tracking throughout the session lifecycle.
- ▶ Implemented bidirectional, event-driven communication using Socket.IO — handling booking requests, provider acceptance/rejection, real-time ride/service state transitions (requested → accepted → en-route → completed), and in-app messaging between users and providers.
- ▶ Built live GPS location tracking with continuous coordinate emission from provider to user, enabling real-time map updates and ETA estimation based on distance calculations.
- ▶ Leveraged Redis for high-performance session management, socket state persistence, and pub/sub-based horizontal scaling — ensuring socket events are correctly routed across multiple server instances under concurrent load.
- ▶ Engineered provider matching logic to efficiently query and rank nearby available providers based on geolocation, service type, and availability status using optimized PostgreSQL spatial queries.
- ▶ Deployed the full backend stack on AWS EC2 behind an Nginx reverse proxy with WebSocket upgrade support configured — enabling persistent socket connections in a production environment.

EDUCATION

B.Tech – Computer Science & Engineering • [RGUKT IIIT Srikakulam](#)

June 2021 – May 2025

Rajiv Gandhi University of Knowledge Technologies | CGPA: 8.2 / 10.0

- ▶ Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, OOP

Pre-University Course (PUC / Intermediate) • [Board of Intermediate Education](#)

Aug 2019 – Apr 2021

Andhra Pradesh | CGPA: 8.9 / 10.0

CERTIFICATIONS

- ▶ **SQL Certification** — Great Learning | 2024
- ▶ **MongoDB Certification** — MongoDB University (Official) | 2023